Christian Hong

Curriculum Vitae

Department of Mathematics, South Hall Room 6607
University of California, Santa Barbara
Santa Barbara, California 93106-3080
christianhong@math.ucsb.edu
hongchristian.github.io

Education

2019–2026 **University of California, Santa Barbara**, *Santa Barbara, California, USA*
Degree: PhD in Applied Mathematics
Advisor: Gustavo Ponce and Hector Ceniceros

2019–2021 **University of California, Santa Barbara**, *Santa Barbara, California, USA*
Degree: Masters of Arts in Applied Mathematics

2014–2018 **University of California, Berkeley**, *Berkeley, California, USA*
Degree: Bachelor of Arts in Applied Mathematics

Academic Papers

C. Hong, *On local energy decay for solutions to the b-family of peakon equations*, Nonlinear Analysis, 253 (2025), 113739
https://www.sciencedirect.com/science/article/pii/S0362546X2400258X

C. Hong, G. Ponce, *On Special Properties of Solutions to the Benjamin-Bona-Mahony Equation*, Journal of Differential Equations, 393 (2024), 321-342
https://www.sciencedirect.com/science/article/pii/S0022039624000913

C. Hong, F. Linares, G. Ponce, *On special properties of solutions to Camassa-Holm equation and related models*, Advanced Studies in Pure Mathematics, forthcoming, https://arxiv.org/abs/2405.08258

A. Sabater, W. Sheppard, C. Hong, Y. Xuan, H. Ceniceros, G. Fredrickson, *Automating the Discovery and Stability of Bulk Morphologies in $A(BA')3$ Miktoarm Star Copolymers*, J. Chem. Phys. 165, 154903 (2026), https://pubs.aip.org/aip/jcp/article-abstract/165/15/154903/3406572/Data-driven-discovery-and-stability-of-bulk

A. Sabater, W. Sheppard, C. Hong, G. Edelen, Y. Xuan, H. Ceniceros, G. Fredrickson, *Variational Autoencoders in the Application of Phase Discovery for the AB Diblock System*, in preparation

Conference Presentations

June 2024 **"On Special Properties of Solutions to the Benjamin-Bona-Mahony Equation"**, *Minisymposium -Evolution Equations and Integrable Systems*, NWCS 2024

July 2026 **"On Special Properties of Solutions to the Benjamin-Bona-Mahony Equation"**, *Special Session -Evolution Equations and Integrable Systems*, AIMS 2026

Seminar Presentations

September 2026 **"On Special Properties of Solutions to the Benjamin-Bona-Mahony Equation"**, *Partial differential equations seminar at ASU*

April 2023 **"Local Well-Posedness of Perturbations to Black Soliton Solutions"**, *SIAM Student Chapter at UCSB*

January 2023 **"An Overview of the Camassa-Holm Equation"**, *SIAM Student Chapter at UCSB*

Conferences Attended

June 2024 **SIAM Conference on Nonlinear Waves and Coherent Structures**, *NWCS 2024*, Lord Baltimore Hotel, Baltimore, MD

July 2025 **Optimal Transport Graduate Summer School**, *University of Santa Barbara*, Santa Barbara, California

July 2026 **The 15th AIMS Conference**, *AIMS 2026*, National and Kapodistrian University of Athens, Athens, Greece

Teaching

August 2024 **University of California, Santa Barbara, Department of Mathematics**, *Santa Barbara, California, USA*

Teaching Associate

Summer 2024: Math 6B - Vector Calculus 2

August 2022 **University of California, Santa Barbara, Department of Mathematics**, *Santa Barbara, California, USA*

Teaching Associate

Summer 2022: Math 6B - Vector Calculus 2

2025-26 **Mentor for Directed Reading Program**

2022-24 **Mentor for Directed Reading Program**

September 2019 - present **University of California, Santa Barbara, Department of Mathematics**, *Santa Barbara, California, USA*

Teaching Assistant

Fall 2019: Math 34A - Calculus for Social and Life Sciences
Winter 2020: Math 4B - Differential Equations
Spring 2020: Math 3B - Calculus 2
Fall 2020: Math 117 - Methods of Analysis
Winter 2021: Math 118B - Introduction to Real Analysis
Spring 2021: Math 118C - Introduction to Real Analysis
Fall 2021: Math 4A - Linear Algebra with Applications
Winter 2022: Math 122A - Introduction to Theory of Complex Variables
Spring 2022: Math 117 - Methods of Analysis
Fall 2022: Math 118A - Introduction to Real Analysis
Winter 2023: Math 6A - Vector Calculus 1
Spring 2023: Math 6B - Vector Calculus 2
Fall 2023: Math 119A - Ordinary Differential Equations
Winter 2024: Math 118B - Introduction to Real Analysis
Spring 2024: Math 6A - Vector Calculus 1
Fall 2024: Math 117 - Methods of Analysis
Winter 2025: Math 124A - Partial Differential Equations
Spring 2025: Math 6B - Vector Calculus 2
Fall 2025: Math 119A - Ordinary Differential Equations
Winter 2026: Math 124A - Partial Differential Equations
Spring 2026: Math 4B - Differential Equations

September 2019 - present **University of California, Santa Barbara, Department of Mathematics**, *Santa Barbara, California, USA*

Summer Teaching Assistant

- Summer 2020: Math 4A - Linear Algebra with Applications
- Summer 2021: Math 6A - Vector Calculus 1
- Summer 2025: Math 119B - Chaotic Dynamics and Bifurcation Theory
- Summer 2025: Math 124B - Fourier Series and Numerical Methods

Academic Employment

September 2019 - present **Teaching Assistant**, *Department of Mathematics*, University of California, Santa Barbara

Awards and Grants

June 2024 **UCSB Doctoral Student Travel Grant**, *Amount: $ 900*

June 2024 **SIAM Travel Award**, *Amount: $ 650*

July 2026 **UCSB Doctoral Student Travel Grant**, *Amount: $ 1500*

Service

June 2023 - March 2025 **UCSB Student Chapter of SIAM**, *Treasurer*

Professional Affiliations

Society for Industrial And Applied Mathematics